PROGRAMMING IN JAVA

Assignment 4

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which access modifier allows access only inside the same package?

a. **default**

b. **private**

c. **public**

d. **protected**

Correct Answer: a

a. **default**

Detailed Solution:

The default access modifier allows access within the same package only.
It is applied when no access modifier is written.
Revise package-level access rules carefully.

QUESTION 2:

Which constructor is called first when an object of a child class is created? The method will be accessible.

a. **Child class constructor**

b. **Parent class constructor**

c. **Interface constructor**

d. **Package constructor**

Correct Answer: b

b. **Parent class constructor**

Detailed Solution:

The parent class constructor is invoked first automatically.
This happens through an implicit call to super().
Revisit inheritance basics and constructor order.

QUESTION 3:

What will be the output of the following code?

```
class Person {
    int a = 1;
    int b = 0;
    public Person() {
        System.out.println(a + b + " Java " );
    }
}

class Employee extends Person {
    int a = 0;
    int b = 1;
    public Employee() {
        System.out.println(""+a + b + " Java " );
    }
}

public class Question {
    public static void main(String args[]) {
        Person p = new Employee();
    }
}
```

a. **1 Java 10**
   0 Java 0

b. **1 Java**
   0 Java

c. **1 Java**
   01 Java

d. **1 java 1**
   0 Java 0

Correct Answer: c

c. **1 Java**
   01 Java

Detailed Solution:

If no super() or this() is included explicitly within the derived class constructor, the super() is implicitly invoked by the compiler. Therefore, in this case, the Person class constructor is called first and then the Employee class constructor is called. Up casting is allowed.

QUESTION 4:

Which statement is used to use a class from another package?

a. **package**

b. **include**

c. **import**

d. **extends**

Correct Answer: c

c. **import**

Detailed Solution:

The import statement allows access to classes from other packages.
It improves code readability and reuse.
Practice basic import statements.

QUESTION 5:

What keyword is used to define a package in Java?

a. **define**

b. **package**

c. **import**

d. **namespace**

Correct Answer: b

b. **package**

Detailed Solution:

The `package` keyword is used in Java to define a package. For example, `package mypackage;` defines a package named `mypackage`.

QUESTION 6:

How do you import a specific class from a package in Java?

a. **import package.*;**

b. **import package.ClassName;**

c. **include package.ClassName;**

d. **use package.ClassName;**

Correct Answer: b

b. **import package.ClassName;**

Detailed Solution:

The syntax `import package.ClassName;` imports a specific class from a package. For example, `import java.util.Scanner;` imports the `Scanner` class from the `java.util` package.

QUESTION 7:

Consider the following 2 programs:

```
// Main1.java ------------------
public class Main1{
    public static void main(String args[]){
        int number = 10;
        System.out.println(number++ + ++number);
    }
}
```

```
// Main2.java ------------------
public class Main2{
    public static void main(String args[]){
        int number = 10;
        System.out.println(++number + number++);
    }
}
```

Choose the best option among the following for the code snippet given above

a. **Both pre-increment and post-increment operators becomes pre-increment during print.**

b. **Both pre-increment and post-increment operators becomes post-increment during print.**

c. **Both Main1 and Main2 classes give the same output.**

d. **Pre-increment and post-increment operators don't work during print.**

Correct Answer: c

c. **Both Main1 and Main2 classes give the same output.**

Detailed Solution:

The output of both the program are 22. Therefore, option **c** is correct and we can eliminate option **d** that the operators don't work. Further, the operators are doing exactly what they are supposed to do i.e. pre-increment first increases the values and post-increment increases the value during the next operation. The print statement is the next operation; hence it received the post incremented value as well making option **a** and **b** invalid.

QUESTION 8:

Consider the code given below.

```
// Teacher.java ------------------
package nptel1;
public class Teacher {
  protected void showMarks() {
    System.out.println("100 Marks");
  }
}
```

```
// Student.java ------------------
package nptel2;
import nptel1.*;
public class Student extends Teacher {
  void show() {
    showMarks();
  }
  public static void main(String[] args) {
    Student st1 = new Student();
    st1.show();
  }
}
```

What is the output of the above Java Code Snippet with `protected` access modifier?

a. **100 Marks**

b. **No output**

c. **Compiler error**

d. **None of the above**

Correct Answer: a

a. **100 Marks**

Detailed Solution:

Through inheritance, one can access a `protected` variable or method of a class even from outside the package. Here, we accessed `Teacher` class of `nptel1` from `Student` class of `nptel2`.

QUESTION 9:

What will happen if you do not specify an access modifier for a class in a package?

a. **The class will be `private` by default.**

b. **The class will be `protected` by default.**

c. **The class will be accessible only within the same package.**

d. **The class will be `public` by default.**

Correct Answer: c

c. The class will be accessible only within the same package.

Detailed Solution:

If no access modifier is specified for a class, it gets the default access level, making it accessible only within the same package.

QUESTION 10:

Which access modifier provides the most restrictive access in Java?

a. **default**

b. **protected**

c. **private**

d. **public**

Correct Answer: c

c. **private**

Detailed Solution:

The `private` modifier provides the most restrictive access, allowing access only within the class where it is defined.